Measure current in one branch

Task 4.4 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Disconnect power before opening a low-voltage loop. Put the configured ammeter in series at the break. Have the teacher check the range, port and wiring before power is applied.

Trace the closed lamp path. Identify where you will break the branch and what unit you expect.

Meter location	Range / units	Expected reading	Teacher check
Point A			
Point B			

02 Plan it on paper

Draw the meter in series. Have a partner and teacher check the drawing and connection before energizing.

My prediction, chosen feature or design limit

Measure current in one branch

Task 4.4 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Take a current reading. Switch power off, move the meter to a second point in the same unbranched loop, then measure again. Compare readings and their precision. Investigate a large difference by checking range, battery and contacts.

Location	Reading and unit	Precision	Comparison or contact check
A			
B			
Repeat			

04 Use the reading

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Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Where did you insert the meter and why? Report two readings and what the comparison shows.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.